

# KRISTHIAN O. PINILI

**Data Analyst · ML Engineer · AWS Certified Machine Learning Specialist**

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## PROFESSIONAL SUMMARY

Computer Science graduate with 4 years of experience in data analysis, machine learning pipeline development, and statistical validation. Engineered a full IoT health data pipeline — from raw 25 Hz PPG signal capture, through Butterworth bandpass filtering and 16-feature extraction (HRV, morphology, frequency-domain), to MSRF model inference and cloud storage — across a medical-grade application. Developed ASTRA, a decision-tree sorting framework validated across 10<sup>6</sup> data elements with ANOVA and Kruskal-Wallis testing achieving a 1.98x average speedup and 8,948x peak gain. AWS Certified Machine Learning Specialist with a well-grounded understanding of data preprocessing, statistical analysis, ML model deployment, and cloud data infrastructure.

## PROFESSIONAL EXPERIENCE

### Full Stack Developer Intern | National Transmission Corporation Jun – Aug 2025

- Analyzed and restructured departmental data workflows — migrated fragmented email-based records to a centralized data system, reducing redundant data handling by ~60% and enabling real-time data retrieval across 5+ divisions.
- Designed data routing logic for an IT/IS Ticketing System, improving incident resolution efficiency by 25% through structured data categorization and automated processing that cut manual handling time by 40%.
- Monitored system data integrity and performance post-deployment, maintaining 99%+ uptime and collaborating with senior engineers to validate data accuracy across all integrated departments.

### QA Engineer | Kill Boring Dead Apr – Jun 2025

- Analyzed and tracked defect data across multiple concurrent client software products, engineering structured test case datasets that achieved 100% critical issue detection and reduced post-launch anomalies by 30%.
- Applied statistical thinking to defect triage and root cause analysis — identifying patterns in pre-release data that enabled proactive resolution before deployment, consistently meeting 100% of quality benchmarks.

### Casting Agent & Digital Coordinator | ModelManagement.com Jan – Jun 2025

- Collected and analyzed talent profile data for 50+ artists, using structured intake data to match talent profiles to project requirements — expanding the platform's data network by 35% through systematic data-driven recruitment.

## NOTABLE PROJECTS

### ASTRA: Adaptive Sorting with Tree-Based Algorithm Selection

Thesis · Lead Architect & Data Analyst · 2025

- Architected an O(n) decision-tree framework that analyzes real-time data biases — entropy, presortedness, and distribution patterns — across 10<sup>6</sup> data elements to dynamically select optimal sorting algorithms, delivering a 1.98x average speedup and peak performance gains of 8,948x over std::sort.
- Validated results using ANOVA and Kruskal-Wallis statistical testing across multiple dataset categories, outperforming black-box ML models through transparent, zero-training logic.

### StrokeSense — IoT PPG Device with ML-Based Stroke Risk Detection

Lead Data Pipeline & Frontend Developer · 2026

- Engineered a complete medical data pipeline: ESP32 + MAX30102 captures raw IR/Red signals at 25 Hz → IBI artifact filtering → overlapping 30-second windows → Butterworth bandpass filter (0.5–8.0 Hz) → SNR/flatline quality gates → 16-feature extraction (HR, SDNN, RMSSD, pNN50, CV\_IBI, PPG\_Amp, RiseTime, FallTime, Width, Area, SampEn, LF, HF, LFHF, SNR, MotionVM).

- Deployed an MSRF (Markov-Switching Random Forest) model combining HMM latent state modeling (Baseline, Transitional, High-Risk) with Random Forest emission classification — processing a 26-feature vector scaled to 19 selected features, validated on a 45-patient clinical dataset with CHA<sub>2</sub>DS<sub>2</sub>-VASc risk labeling.
- Managed a Convex cloud database with 24 relational tables following a NoSQL-over-SQL strategy — processing ~180 HTTP requests per 60-second session and storing structured feature CSVs with full cloud upload/retrieval.

## TECHNICAL SKILLS

**Data Analysis & ML:** Statistical analysis (ANOVA, Kruskal-Wallis) · Feature engineering · ML model training & deployment · Signal processing · HRV analysis · Markov-Switching Random Forest (MSRF)

**Data Processing:** Python (SciPy, NumPy, Pandas) · Butterworth bandpass filtering · Welch PSD estimation · CSV pipeline engineering · Window-based analysis · SNR calculation

**Cloud & Storage:** AWS Certified ML Specialty · AWS: EC2, S3, Lambda · Convex cloud database · IaaS / PaaS / SaaS · NoSQL-over-SQL · Cloud file storage

**Databases & Tools:** SQL / SQLite · Relational schema design (24-table medical DB) · FastAPI · REST APIs · Git & GitHub · Dart / Flutter · JavaScript

**Certifications:** AWS Certified Machine Learning Specialty · Cisco Certified · CSS NCII — Best in Computer System Servicing

## EDUCATION

**BS Computer Science** | Batangas State University, Batangas City 2022 – 2026 (*Expected*)

*Best in Computer System Servicing (NCII) · Cisco Certified · AWS Certified ML Specialist · Thesis: ASTRA (8,948x peak speedup)*